

Shabab Ahmed

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PROFESSIONAL PROFILE

Ph.D. Candidate in Economics with expertise in structural modeling, contract theory, and applied econometrics. Experienced in developing dynamic models, analyzing large-scale survey data, and translating research into insights that inform business and policy decisions.

EDUCATION

Ph.D. in Economics, University of Washington: Expected **November 2026**

- Dissertation:
 - Dynamic moral hazard and incentive design**: how institutions should adapt when preferences evolve over time, applied to NFL draft policy
 - High-dimensional demand estimation**: using Lasso and Elastic Net regularization with debiasing corrections to estimate elasticity under collinearity, validated via Monte Carlo simulation
 - Dynamic moral hazard under a fixed budget**: how agents allocate spending across time under a budget constraint, and how this can explain patterns like budget hoarding and end-of-year spending spikes
- Passed general examination; presented research in departmental seminars.

M.A. in Economics, University of Washington: Dec 2022 — GPA 3.89

M.S. in Applied Mathematics, University of Washington: Jun 2020 — GPA 3.87

B.A. in Mathematics and Economics, Colby College: May 2019 — GPA 3.93

- Magna Cum Laude, Phi Beta Kappa, Distinction in Economics & Mathematics, Dean's List (6 semesters).

EXPERIENCE

Economist Intern, Amazon, Bellevue, WA (*Jul 2025 – Oct 2025*)

- Built structural demand models on large-scale survey data to estimate customer preferences for Project Kuiper, with findings that helped inform the team's pricing strategy.
- Used a tree-based feature-importance model to identify the key drivers of customer response to determine which inputs entered the structural model.
- Built a simulation platform to evaluate counterfactual business scenarios across multiple pricing and product configurations.
- Authored and presented a technical and business report that translated econometric findings into recommendations for senior stakeholders.

Graduate Teaching Assistant & Independent Instructor, University of Washington (*2019 – Present*)

- Independent Instructor for Game Theory, with full responsibility for course delivery, assignments, and assessment.
- Teaching Assistant for Graduate Microeconomic Theory and undergraduate courses, leading discussion sections and review sessions.

SELECTED PROJECTS

NBA Decade Comparison: PCA and Machine Learning Analysis — Winter 2020

- Built classifiers on 13,000 player-season observations (~95 features scraped from Basketball Reference). Used PCA and LDA to predict player position and decade, and L1-regularized logistic regression to predict Win Shares quartile.
- Achieved 72.6% accuracy on the Win Shares model, validated via 50-fold cross-validation. Raw features outperformed PCA-compressed inputs by over 25 points, which shaped the final feature strategy.

Game Theory in Cricket & Attacks on RSA — 2019

- Built a formal game-theoretic model of bowler-batsman strategy, validated against an original dataset built from match commentary.
- Implemented and analyzed multiple RSA cryptographic attacks in Python.

TECHNICAL SKILLS

Methods: Structural demand estimation, dynamic optimization, Lasso/Elastic Net regularization, debiased ML, causal inference (IV, DiD, panel data), PCA, LDA, logistic regression, tree-based models

Languages & Tools: Python, R, Stata, SQL, MATLAB, Sage, Mathematica, LaTeX, Qualtrics, Excel

Languages: English (fluent), Bengali (fluent), Hindi (conversational)